INCH-POUND
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29 September 2009
SUPERSEDING
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DETAIL SPECIFICATION

REGULATOR, 650 AMP, SOLID STATE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a solid state voltage regulator for use in engine driven military ground vehicles.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

Comments, suggestions, or questions on this document should be addressed to U.S. Army Tank-automotive and Armaments Command, ATTN: RDTA-EN/DM/STND, MS# 268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to DAMI_STANDARDIZATION@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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SPECIFICATIONS

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-62186 - Generator, Alternating Current (650 Ampere Rectified), 28 Volts Direct Current (Oil Cooled)

STANDARDS

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-461 - Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment (See 4.2.1)
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines (See 4.2.1)

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Service, Building 4/D, 700 Robbins Avenue, Philadelphia, PA 19111-5094 or website: http://assist.daps.dla.mil/quicksearch/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

DRAWINGS

ARMY

11655469 - Generator AC/DC (Oil Cooled)
12354334 - Regulator, Solid State.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, RDTA-EN, 6501 E. 11 Mile Road, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

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GENERAL MOTORS CORPORATION (GM)

GM 9540P - Accelerated Corrosion Test.

(Copies of GM publications may be obtained from General Motors Corporation, c/o Global Engineering, 15 Inverness Way, Englewood, CO 80112 or at website: http://www.ihs.com/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2 and 6.3), a sample shall be subjected to first article inspection in accordance with 4.1.1.

3.2 Design, materials, and manufacturing processes. The design, materials, and manufacturing process shall conform to Drawing 12354334.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs (see 6.5.1).

3.3 Operating requirements. All regulators shall meet the performance requirements as specified in 3.3.1 through 3.3.11 when operating on a generator-alternator conforming to MIL-DTL-62186 (see 4.4.1).

3.3.1 Voltage regulation. The regulator shall establish and maintain output voltage in accordance with figures 1 and 2, throughout the generator-alternator speed range of 2400 to 8000 revolutions per minute (rpm) (see 4.4.1.1).

3.3.2 Ripple voltage. Ripple voltage shall not exceed 4 volts (V) peak-to-peak, with output voltage at 28 V direct current (dc), throughout the generator speed range of 2400 to 8000 rpm (see 4.4.1.2).

3.3.3 Overvoltage. The regulator shall deactivate the field circuit when the system voltage (E_s) reaches 33.0 ± 1.0 V. Deactivation and return to battery voltage shall be accomplished in 0.07 seconds. Reset shall be manual and shall be located on the regulator (4.4.1.3).

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3.3.4 False trips. False tripping of the regulator shall not be caused by buildup, load switching, and input speed acceleration or deceleration (see 4.4.1).

3.3.5 Current limit. The regulator shall limit the generator-alternator output current in accordance with figure 3. Stable current limiting shall be established within 0.50 seconds. System voltage undershoot shall be not below 18 V (see 4.4.1.4).

3.3.6 Accessory control. The regulator shall incorporate circuitry to energize an accessory control device that closes when the generator-alternator speed is between 900 and 1100 rpm. During shutdown, this device shall open before the generator-alternator speed reaches 800 rpm. The circuitry, when energized, shall produce at least 0.25 amperes (A) at a voltage not less than 18 V (see 4.4.1.5).

3.3.7 Fault protection. The regulator shall include circuitry that deactivates the field circuit in the event of wiring malfunctions. This circuitry shall operate only if a malfunction in any of the trip types of table I creates a condition which would damage the regulator or associated electrical power sources. Maximum voltage and trip time limits shall be as specified in table I for the three trip types (see 4.4.1.6).

TABLE I. Trip types.

Trip type	Generator-alternator rpm	Alternator output voltage (E_A) V	System voltage (E_S) V	Time (seconds)
Regulator sensing	2400	35.0	35.0	1.00
Input lead	8000	35.0	35.0	0.4
Alternator positive	2400	40.0	35.0	1.00
Output lead	8000	115.0	35.0	0.4
Alternator negative	2400	40.0	35.0	1.00
Output lead	8000	115.0	35.0	0.4

3.3.8 Dielectric strength. The regulator circuit insulation to ground shall not permit a leakage current greater than 5 milliamperes (mA) with 500 V alternating current (ac) simultaneously applied between all receptacle pins and the case, and with the case ground lead disconnected (see 4.4.1.7).

3.3.9 Short circuit protection. Regulator circuitry shall prevent damage to internal components in the event of a short circuit from pin “E” or “F” (see 3.4.3) to ground during normal regulator operation (see 4.4.1.8).

3.3.10 Electromagnetic interference (EMI). The regulator shall not exceed the broadband emission limits as specified in figures 4 and 5 (4.4.1.9).

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3.3.11 Reverse polarity protection. The regulator circuit shall not be damaged in the event reverse polarity is applied to pins "A" and "C" (see 3.4.3 and 4.4.1.10).

3.4 Interface requirements. Each regulator shall accommodate the inputs, interfaces, and envelope requirements as specified in 3.4.1 through 3.4.4 (see 4.4.2).

3.4.1 Envelope and mounting dimensions. The regulator shall conform to the envelope and mounting dimensions as specified on Drawing 12354334 (see 4.4.2).

3.4.2 Mating. The regulator shall mate with the shell socket insert arrangement as specified on Drawing 12354334. Contacts shall remain positively retained in the insert arrangement throughout all conditions of intended use without causing damage or compromising performance (see 4.4.2).

3.4.3 Electrical interface. Electrical connector pin designations to the utilizing vehicle shall be provided as specified in table II (see 4.4.2).

TABLE II. Electrical interface.

Connector pin function	Connector pin designations
Regulation sensing input	A
Current limit signal input	B
Ground	C
Generator-alternator AC signal input	D
Generator-alternator field control output	E
Accessory control device signal output	F
Overvoltage test circuit input	G

3.4.4 Weight. The regulator shall not exceed 6 pounds (lb) (2.7 kilograms (kg)) (see 4.4.2).

3.5 Support and ownership requirements. The regulator shall possess the life cycle ownership characteristics as specified in 3.5.1 through 3.5.5 (see 4.4.3).

3.5.1 Dissimilar metals. Except where necessary to complete an electrical circuit, provisions shall be made to prevent galvanic corrosion ascribable to contact between dissimilar metals (see 4.4.3.1).

3.5.2 External surfaces. All external surfaces withstand application of a 1 percent (%) salt mist for a period of at least 192 hours without corroding (see 4.4.3.2).

3.5.3 Endurance. Regulator performance shall not be compromised during and after 3000 hours of operation as specified in table III (see 4.4.3.3).

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TABLE III. Regulator endurance time schedule and conditions.

1/ Time, hours	2/ RPM	3/ System voltage, volts	Alternator resistive load			Load cycles, minutes	Cyclic load, seconds	
			Constant current, A	Cyclic current, A	Total current, A		On	Off
1.5	2400	26.5 ±0.75	320	330	650	90	3	57
7.5	5120	26.5 ±0.75	320	330	650	450	3	57
14.5	3520	26.5 ±0.75	320	330	650	870	3	57
0.5	8000	26.5 ±0.75	650	0	650	30	--	--

1/ 24 hours per cycle.
2/ Alternator.
3/ System voltage shall be applied at +225 degrees Fahrenheit (°F).

3.5.4 Safety. The regulator case shall not pose an electrical hazard due to improper grounding (see 4.4.3.4).

3.5.5 Nameplate. A permanently and legibly marked nameplate shall be conspicuously attached to the regulator housing and shall contain, as a minimum, the following information (see 4.4.3.5):

a. Regulator, solid state.
b. 28 volts dc.
c. Army part number.
d. National stock number.
e. Manufacturer's identification (CAGE).
f. Manufacturer's serial number.
g. Contract number.
h. Date of manufacture.
i. US.

3.6 Operating environment requirements. Each regulator shall operate under the environmental conditions as specified in 3.6.1 through 3.6.6 (see 4.4.4).

3.6.1 Shock. The regulator shall withstand 40 gravity units (g) of sawtooth shock pulses in accordance with figure 6 for an 18 ± 3 millisecond (ms) duration in each of the major mutually perpendicular axes without damage (see 4.4.4.1).

3.6.2 Vibration. The regulator shall withstand a maximum input vibration as specified in figures 7 and 8 in each of three mutually perpendicular axes without damage (see 4.4.4.2).

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3.6.3 Waterproofness. When submerged in salt water, the regulator shall withstand 10 pounds per square inch gage (psig) (69 kilopascals (kPa)) internal pressure and internal vacuum without leaking (see 4.4.4.3).

3.6.4 Extreme temperature. The regulator shall operate at ambient temperatures in the range of -65 to 225°F (-54 to 107°C) without a compromise in performance. The range of regulated voltage over the total operational temperature range shall conform to figure 2 (see 4.4.4.4).

3.6.5 Humidity. The regulator shall show no visible damage, corrosion, or swelling, nor sustain a loss in performance due to exposure to 5 continuous 48 hour humidity cycles in accordance with figure 9 (see 4.4.4.5).

3.6.6 Fungus. The regulator shall operate in environments conducive to spore germination and growth (air temperature between 75 to 88 degrees Fahrenheit (°F) (24 to 31 degrees Celsius (°C)) and relative humidity between 90 to 100 percent (%)) without a compromise in performance attributable to the deleterious effects of fungus (see 4.4.4.6).

4. VERIFICATION

4.1 Classification of inspections. Inspection requirements are classified as follows:

a. First article inspection (see 4.1.1).
b. Conformance inspection (see 4.1.2).

4.1.1 First article inspection. When specified (see 6.2 and 6.3), a first article sample shall be selected in accordance with 4.1.1.1 and 4.1.1.2 and shall be subject to all of the comprehensive verifications as specified in table IV.

4.1.1.1 Preproduction inspection. When specified (see 6.2 and 6.3), a preproduction sample shall consist of two regulators.

4.1.1.2 Initial production inspection. Unless otherwise specified (see 6.2 and 6.3), an initial production sample shall consist of two regulators from the first ten produced.

4.1.2 Conformance inspection. As specified in the contract (see 6.2 and 6.4), a 100% conformance inspection shall be conducted and shall include the following verifications: All the abbreviated tests, and the accessory control, dielectric strength, short circuit protection, reverse polarity, safety, and waterproofness verifications, as specified in table IV.

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TABLE IV. Verification methods.

Title	Require-ment	Verifi-cation 1/	Title	Require-ment	Verifi-caiton 1/
Operating requirements	3.3	4.4.1	Mating	3.4.2	4.4.2
Voltage regulation	3.3.1	4.4.1.1,	Electrical interface	3.4.3	4.4.2
		*4.4.1.1.1	Weight	3.4.4	4.4.2
Ripple voltage	3.3.2	4.4.1.2,	**Support and ownership**	3.5	4.4.3
		*4.4.1.2.1	**requirements**		
Overvoltage	3.3.3	4.4.1.3	Dissimilar metals	3.5.1	4.4.3.1
		*4.4.1.3.1	External surfaces	3.5.2	4.4.3.2
False trips	3.3.4	4.4.1	Endurance	3.5.3	4.4.3.3
Current limit	3.3.5	4.4.1.4,	Safety	3.5.4	4.4.3.4
		*4.4.1.4.1	Nameplate	3.5.5	4.4.3.5
Accessory control	3.3.6	4.4.1.5	**Operating environment**	3.6	4.4.4
Fault protection	3.3.7	4.4.1.6,	**requirements**		
		*4.4.1.6.1	Shock	3.6.1	4.4.4.1
Dielectric strength	3.3.8	4.4.1.7	Vibration	3.6.2	4.4.4.2
Short circuit protection	3.3.9	4.4.1.8	Waterproofness	3.6.3	4.4.4.3
EMI	3.3.10	4.4.1.9			*4.4.3.1
Reverse polarity	3.3.11	4.4.1.10	Extreme temperature	3.6.4	4.4.4.4
Interface requirements	3.4	4.4.2	Humidity	3.6.5	4.4.4.5
Envelope and mounting	3.4.1	4.4.2	Fungus	3.6.6	4.4.4.6

1/ All verifications shall be considered as comprehensive unless denoted by an asterisk (*). Verifications denoted by an asterisk (*) shall be considered abbreviated (see4.1.2).

4.2 Verification methods. The types of verification methods included in this section are visual, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously-approved or previously-qualified designs.

4.2.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost-effective sampling procedures, to verify performance.

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 77 ± 15°F (25 ± 7°C).
b. Barometric pressure: 28.5 (+2, -3) inches (96.5 (+6.8, -10 kPa)) of mercury.
c. Relative humidity: 50 ± 30 %.

4.3.1 Laboratory test conditions. Unless otherwise specified (see 6.2), the laboratory test conditions shall be as follows:

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a. Oscillographics shall have a frequency response from 0 to 2000 Hz which is flat within ± 5%.
b. Prior to each performance test, the regulator shall be operated for 15 minutes at 3000 rpm at 650 A alternator current for temperature stabilization.
c. Verifications under 4.4.1 shall utilize a test circuit as specified in figure 10. Electrical switches SW8, SW9 and SW13 shall be closed at all times unless otherwise specified, and SW3 shall always be open unless otherwise specified. SW1 shall be the first closed when energizing the regulator followed by SW2. SW4 shall be closed whenever resistive load is required during these tests. When de-energizing the system, SW1 shall always be the last switch opened unless otherwise specified. SW1 shall be opened and closed only with the rotary power source at rest.

4.4 Order of inspection. The inspection sequence may be in any order with the following exceptions. The operating environment verifications (see 4.4.4), the accelerated corrosion test (see 4.4.3.2), and the endurance test (see 4.4.3.3) shall precede the operating requirements verifications (see 4.4.1). The fungus verification (see 4.4.4.6) shall precede the corrosion test (see 4.4.3.2).

4.4.1 Operating requirements verifications. Each regulator shall perform as specified in the tests under 4.4.1 using a generator-alternator conforming to MIL-DTL-62186. Unless otherwise specified herein (see 4.4.1.6.1), the regulator shall perform without false tripping. When specified (see 6.2), the regulator shall meet the requirements under 4.4.1 when installed on the vehicle for which item is being procured (see 3.3.4).

4.4.1.1 Voltage regulation verification. To determine conformance to 3.3.1, the regulator shall be installed in a circuit as shown in figure 10 and operated. With SW1 and SW2 closed, resistive loads of zero (battery only), 100A and 650A shall be applied at generator-alternator speeds of 2400, 4000, 6000 and 8000 rpm. The system voltage (E_s) shall be in accordance with figure 2. With a resistive load of 650A suddenly applied and removed (application and release times shall not exceed 10 ms) at generator-alternator speeds of 2400, 4000, 6000 and 8000 rpm, the system voltage (E_s) shall be in accordance with figure 1.

4.4.1.1.1 Abbreviated voltage regulation verification. With the stabilized regulator installed in a circuit as specified in 4.4.1.1, the resistive loads specified in the following procedures shall be applied. The output voltage shall be in accordance with figure 2 (see 4.1.2).

a. Operate generator-alternator at 2400 rpm with resistance load at zero (battery only).
b. With generator-alternator at 2400 rpm, add resistance load to establish a total current flow of 100 A.
c. With generator-alternator speed at 2400 rpm, increase resistive load to establish a current flow of 650 A.

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d. Increase generator-alternator speed to 4000 rpm, and remove the resistance load.
e. With generator-alternator speed at 4000 rpm add resistance load to establish a total current flow of 100 A.
f. With generator-alternator at 4000 rpm, increase resistance load to establish a current flow of 650 A.
g. Increase generator-alternator speed to 6000 rpm and remove the resistance load.
h. With generator-alternator at 6000 rpm, add resistance load to establish a total current flow of 100 A.
i. With generator-alternator speed at 6000 rpm, increase resistive load to establish a current flow of 650 A.
j. Increase generator-alternator speed to 8000 rpm and remove the resistance load.
k. With generator-alternator at 8000 rpm, add resistance load to establish a total current flow of 100 A.
l. With generator-alternator speed at 8000 rpm, increase resistive load to establish a current flow to 650 A.

4.4.1.2 Ripple voltage verification. To determine conformance to 3.3.2, the regulator shall be connected in a circuit as shown in figure 10 and operated at the specified rate and load to produce a current flow as indicated below. Ripple voltage at point E_S shall not exceed 4 V peak-to-peak.

Rpm	Alternator output current, A (I_A)
3000	50
3000	650
8000	50
8000	650

4.4.1.2.1 Abbreviated ripple voltage verification. The regulator shall be tested as specified in 4.4.1.2, except that the test point shall be 8000 rpm at a load producing an alternator output current flow of 650 amperes (I_A) (see 4.1.2).

4.4.1.3 Overvoltage verification. To determine conformance to 3.3.3, the regulator shall be connected in a circuit as shown in figure 10 and shall be operated as follows. With SW1 and SW2 closed, adjust alternator speed to 5000 rpm. Observe system voltage (E_s) on an oscillograph as SW3 is closed. Determine E_S where trip occurs. Open SW3 and reset overvoltage switch on regulator. The field circuit shall be deactivated when the system voltage (E_s) reaches 33.0 ± 1.0 V. Deactivation and return to battery voltage shall be no greater than 0.07 seconds.

4.4.1.3.1 Abbreviated overvoltage verification. With the regulator connected as shown in figure 11, the power supply voltage shall be increased until overvoltage circuit breaker trips. The dc voltmeter reading shall be determined only at the overvoltage switch trip point. The dc

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power shall be disconnected and the regulator overvoltage switch shall be reset. The field circuit shall be deactivated when the system voltage (E_s) reaches 33.0 ± 1.0 V (see 4.1.2).

4.4.1.4 Current limit. To determine conformance to 3.3.5, the regulator shall be installed in a circuit such as shown in figure 10 and the following test shall be performed:

a. Current limit, transient. Open switch SW13 and operate the regulator for 15 minutes with a 500 ampere alternator current flow at 5000 rpm. Disconnect all load from the alternator except batteries. Alternator current (I_A) and system voltage (E_s) shall be observed. Connect a resistive load of 0.028 ohms. Stable current limiting shall be established within 0.15 seconds and system voltage undershoot shall not be below 18 V.
b. Current limit, regulation. Apply a resistive load to produce a current flow of 50 A from the alternator at 2400 rpm. Increase the loading in 50A increments until the system voltage is reduced to less than 24 V. The system voltage and current shall be in accordance with figure 3. Repeat at 5000 rpm alternator speed.

4.4.1.4.1 Abbreviated current limit. With the regulator installed as specified in 4.4.1.4, operationally stabilized, and with the alternator at 2400 rpm, a resistive load shall be applied to produce a current flow of 400 A from the alternator. The load shall be increased until E_s is 26 ± 0.1 V, and I_A shall conform to the limits as specified in figure 3. Repeat at an alternator speed of 5000 rpm (see 4.1.2).

4.4.1.5 Accessory control device verification. To determine conformance to 3.3.6, the regulator shall be tested while it is installed in a circuit as is shown in figure 10. With SW1 and SW2 closed, the alternator speed shall be slowly increased from zero rpm until the accessory control device closes. The accessory control device shall close when the generator-alternator speed is between 900 and 1100 rpm. The alternator speed shall be slowly decreased until the accessory control device opens. The accessory control device shall open before the generator-alternator speed reaches 800 rpm during shutdown. Verify that the circuitry produces a minimum 0.25 A at 18 V.

4.4.1.6 Fault protection. To determine conformance to 3.3.7, the regulator shall be installed in a circuit as shown in figure 10 and the following tests shall be performed.

a. Regulator sensing input lead.
   1. Operate the regulator and alternator at 2400 rpm with no external alternator load with SW10 closed. Set variable resistance R10 at zero ohms and open SW10. Slowly increase R10 until the protective circuit trips. Just before the trip occurs, verify that E_A and E_S do not exceed the limits specified in table I. Leave R10 at the final setting. Close SW10 and reset the regulator overvoltage switch.

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2. Operating at 2400 rpm, open SW10. Just before the trip occurs, verify that E_A, E_S and trip times do not exceed the limits specified in table I. After the circuit trips, close SW10 and reset regulator overvoltage switch.
3. Repeat a1 above at 8000 rpm.
4. Repeat a2 above at 8000 rpm.

b. Alternator positive output cable.
1. Operate the regulator and alternator at 2400 rpm, with no external alternator load with SW11 closed. Set R11 at zero ohms and open SW11. Slowly increase R11 until the protective circuit trips. Just before the trip occurs, verify that E_A and E_S do not exceed the limits specified in table I. Leave R11 at the final setting. Close SW11 and reset overvoltage switch.
2. Operating at 2400 rpm, open SW11. Just before the trip occurs, verify that E_A, E_S and trip times do not exceed the limits specified in table I. After the circuit trips, close SW11 and reset the overvoltage switch.
3. Repeat b1 above at 8000 rpm.
4. Repeat b2 above at 8000 rpm.

c. Alternator negative output cable.
1. Operate the regulator and alternator at 2400 rpm with no external alternator load with SW12 closed. Set R12 at zero ohms and open SW12. Slowly increase R12 until the protective circuit trips. Just before the trip occurs, verify that E_A and E_S do not exceed the limits specified in table I. Leave R12 at the final setting. Close SW12 and reset the overvoltage switch.
2. Operating at 2400 rpm, open SW12. Just before the trip occurs, verify that E_A, E_S and trip times do not exceed the limits specified in table I. After the circuit trips, close SW12 and reset the overvoltage switch.
3. Repeat c1 above at 8000 rpm.
4. Repeat c2 above at 8000 rpm.

4.4.1.6.1 Abbreviated fault protection circuitry. Operate the regulator with no external alternator load. The overvoltage switch shall trip after steps 1 and 2 are performed (see 4.1.2).

Step 1. Open SW8.
Step 2. Close SW8, reset overvoltage switch and open SW9.
Step 3. Close SW9 and reset the overvoltage switch. The regulator shall operate without tripping.

4.4.1.7 Dielectric strength verification. To determine conformance to 3.3.8, remove the cover and disconnect the internal circuit ground lead from the case. Connect all the pins of the regulator receptacle together. Starting with the tester voltage at 0 V, slowly increase the voltage to 500 V ac for 1 minute and then decrease the tester voltage to zero before removing the test leads and verify that the resistance is greater than 100 kilo ohms.

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4.4.1.8 Short circuit protection verification. To determine conformance to 3.3.9, the regulator shall be connected to the test configuration as shown in figure 12. A test shall be considered successful when table V is followed in the order shown with the resulting circuit breaker and lamp (L1) response. The regulator shall then successfully pass the test specified in 4.4.1.1.1 to verify that no circuit has been damaged.

TABLE V. Short circuit protection.

SW1	SW2	Circuit Breaker 1/	L1
Open	Open	On	On
Close	Open	Off	Off
Open	Open	Off	Off
Open	Open	On	On
Open	Close	Off	Off
Open	Open	Off	Off
Open	Open	On	On

1/ Breaker will automatically trip off, but must be reset by switching on by hand.

4.4.1.9 EMI verification. To determine conformance to 3.3.10, the regulator shall be subjected to EMI tests for radiated and conducted broadband emissions in accordance with MIL-STD-461, methods CE102 and RE102, or equivalent (see 4.2.1). The regulator shall be within the limits as specified in figures 4 and 5.

4.4.1.10 Reverse polarity protection verification. To determine conformance to 3.3.11, with the regulator circuit breaker switched on, 24 V dc shall be applied to pin C and the ground to pin A for 1 minute. The regulator shall meet the performance test under 4.4.1.1.1 after applying reverse polarity voltage to verify that no circuits have been damaged.

4.4.2 Interface requirements verifications. To determine conformance to 3.4, the regulator shall be visually, functionally, and equipmentally examined to verify that envelope and mounting dimensions are in accordance with Drawing 12354334. When specified (see 6.2), conformance shall be demonstrated when installed on the vehicle for which the item is being procured.

4.4.3 Support and ownership verifications. To determine conformance to 3.5, complete each verification under 4.4.3.

4.4.3.1 Dissimilar metals verification. To determine conformance to 3.5.1, visually verify that dissimilar metals are separated.

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4.4.3.2 External surfaces verification. To determine conformance to 3.5.2, the regulator shall then be tested in accordance with GM 9540P, Method B, except that the duration shall be for four test cycles (192 hours total). There shall be no evidence of corrosion.

4.4.3.3 Endurance verification. To determine conformance to 3.5.3, the regulator shall be subjected to 3000 hours of operation in accordance with table II at an ambient temperature of 225 ±5°F, and shall meet the requirements specified therein. The regulator shall require no servicing, adjustment, or replacement of parts. Conformance to the endurance test shall also be verified by subsequently passing the operating requirements verifications as specified in 4.4.1.

4.4.3.4 Safety verification. To determine conformance to 3.5.4, continuity between the regulator case and pin C of the connector shall be less than 1 ohm while the regulator is disconnected from the system.

4.4.3.5 Nameplate verification. To determine conformance to 3.5.5, visually inspect that name plate is accurately, legibly, and permanently marked as follows:

a. Regulator, solid state.
b. 28 volts dc.
c. Army part number.
d. National stock number.
e. Manufacturer's identification (CAGE).
f. Manufacturer's serial number.
g. Contract number.
h. Date of manufacture.
i. US.

4.4.4 Operating environment verifications. To determine conformance to 3.6, complete each verification under 4.4.4. In addition to withstanding the tests as specified under 4.4.4, conformance to operating environment requirements shall be further demonstrated by subsequently passing the operating requirements verifications as specified under 4.4.1.

4.4.4.1 Shock verification. To determine conformance to 3.6.1, the regulator shall be mounted as in use and shall be subjected to a total of 18 shock pulses of 40 g at 18 ± ms duration in each direction along three mutually perpendicular axes. The shock pulses shall be in accordance with the configuration and tolerance limits as specified in figure 6. There shall be no evidence of damage.

4.4.4.2 Vibration verification. To determine conformance to 3.6.2, the regulator shall be attached to the vibration generator. The fixture shall incorporate actual service structures as much as possible to minimize unrealistic response characteristics during test exposures. Vibration shall be applied for 100 minutes in each direction along each of three perpendicular axes. Amplitude shall be 4 gravity units. Frequency shall be 5 to 500 to 5 Hz in a 15 minute

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time sweep. Test level shall be in accordance with the curve of figure 7. Resonant frequencies of the regulator shall be determined and a sinusoidal resonance search shall be conducted for the four most severe resonant frequencies of the unit. The regulator shall be subjected to 16.34 minutes at each severe resonant frequency within each 100 minutes sweep test. The frequency of applied vibration shall be swept over the specified range in accordance with figure 8. The sweep time is that of an ascending plus a descending time shown on figure 8. The vibration resistance test shall be performed at an ambient temperature of -65°F and 225°F (-54 to 107°C). There shall be no evidence of damage.

4.4.4.3 Waterproofness verification. To determine conformance to 3.6.3, the regulator shall be submerged into a salt water bath at 73 ±18°F (23 ±10°C) with a minimum depth of 1 inch (2.54 centimeters (cm)) covering the unit. A 10 psig (69 kPa) vacuum shall be applied and the regulator operated in accordance with 4.4.1.1. The unit shall be removed from the bath and thoroughly dried. Carefully remove the cover and inspect the regulator for water leakage. Evidence of moisture in or on the regulator interior shall constitute a failure of this test. Carefully replace the cover, remove the vacuum pump and connect to a dry nitrogen source. Submerge the regulator as before, apply 10 psig dry nitrogen pressure and test in accordance with 4.4.1.1. Escaping bubbles from the regulator interior shall constitute a failure of this test.

4.4.4.3.1 Abbreviated waterproofness verification. The regulator shall be subjected to and pass the test of 4.4.4.3 except that only a non-operational pressurization test shall be conducted, using compressed air instead of nitrogen (see 4.1.2).

4.4.4.4 Extreme temperature verification. To determine conformance to 3.6.4, the regulator shall be placed in a chamber suitable for the development and maintenance of the specified temperatures specified below.

a. Low temperature. After stabilization in the chamber, the regulator shall be kept at -80°F (-62°C) for 12 hours. After storage at -80°F (-62°C), the temperature shall be raised to -65°F (-54°C). When stabilized at this temperature, the regulator shall be subjected to and pass the test as specified in 4.4.1.1.1, 4.4.1.2.1, 4.4.1.3.1, 4.4.1.4.1, 4.4.1.5, 4.4.1.6.1, 4.4.1.8, 4.4.1.10, and 4.4.3.4.
b. High temperature. With the regulator installed as ready to operate, the internal chamber temperature shall be raised to 160°F (71°C) within 1 hour time period. This temperature shall be maintained for an additional 4 hours. The internal chamber temperature shall then be lowered to 120°F (49°C) within a 1 hour time period. This cycle shall be repeated three times for a total of 36 hours. The internal chamber temperature shall be adjusted to 225°F (107°C) and maintained until the regulator is temperature stabilized. The regulator shall be operated and pass the tests as specified in 4.4.1.1.1, 4.4.1.2.1, 4.4.1.3.1, 4.4.1.4.1, 4.4.1.5, 4.4.1.6.1, 4.4.1.8, 4.4.1.10, and 4.4.3.4. The regulator shall be returned and

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stabilized to standard ambient conditions. No physical damage shall exist when the regulator is operated.

4.4.4.5 Humidity. To determine conformance to 3.6.5, the regulator shall be exposed to the following procedure:

a. The regulator shall be installed as in use in a chamber vented and baffled to assure an even flow of warm humid air as specified herein. No condensation shall be permitted to drip on the regulator. Relative humidity shall be measured by the dry bulb-wet bulb thermometer comparison method, or equivalent. Air velocity in the internal test space shall not exceed 150 feet per minute (ft/min) (45.6 meters per minute (m/min)). Steam or distilled, demineralized or deionized water having a pH value between 6 and 7.2 at 73 ± 3°F (23 ± 1°C) shall be used to obtain the specified humidity. No rust or corrosive contaminants shall be imposed on the test sample by the test facility. During this test, no repair or replacement shall be permitted and the regulator shall be operated only when necessary.
b. After having been dried for 24 hours at 129 ±3°F (54 ±1.5°C), the regulator shall be conditioned at 73 ±3°F and 50 ±10% relative humidity for 24 hours. The internal chamber temperature shall be raised to 86 ±3°F (30 ±2°C) and 94 ±4% relative humidity. The regulator shall be exposed to 5 continuous 48 hour cycles in accordance with figure 9. Accumulated moisture may be removed by turning the regulator upside down or shaking. Wiping shall not be permitted. The regulator shall show no visible damage, corrosion or swelling.

4.4.4.6 Fungus verification. To determine conformance to 3.6.6, the regulator shall be subjected to fungal incubation for 28 days in accordance with method 508.4 of MIL-STD-810, or equivalent (see 4.2.1). Samples shall be incubated under cyclic temperature and humidity conditions and shall include 20 hours of relative humidity at 95 ±5% at an air temperature of 86 ± 2°F (30 ±1°C), followed by 4 hours of 100% relative humidity at 77 ± 2°F (25 ± 1°C). There shall be no evidence of fungal growth.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The regulator is intended to control and monitor the alternator to provide prescribed electrical power for engine driven military surface vehicles. Given that the extreme battlefield situations under which this item must perform would cause catastrophic failure in commercial products (extreme conditions of shock, vibration, temperature, and humidity (see 3.6)), this item is military unique.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. If required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2, and 2.3).
c. If first article testing is other than as specified (see 3.1 and 4.1.1).
d. If preproduction testing is required (see 4.1.1.1).
e. If initial production testing is other than as specified (see 4.1.1.2).
f. Specification of conformance testing (see 4.1.2 and 6.4).
g. If inspection conditions should be other than as specified (see 4.3 and 4.3.1).
h. If trial installation on a military vehicle is required (see 4.4.1 and 4.4.2).
i. Packaging requirements (see 5.1).

6.3 First article. When requiring a first article inspection (see 3.1), contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample (see 4.1.1), a pre production item (see 4.1.1.1), a first production item (see 4.1.1.2), and the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.4 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from table IV (see 4.1.2 and 6.2) and their designated frequency based on a risk assessment for the procurement.

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6.5 Definitions.

6.5.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 3.2.1 and 6.5.2).

6.5.2 Solid waste. “Solid waste” means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.6 Subject term (key word) listing.

Current limit
Direct current
Electrical power
False trips
Fault protection
M60 series tank
Overvoltage
Voltage regulation

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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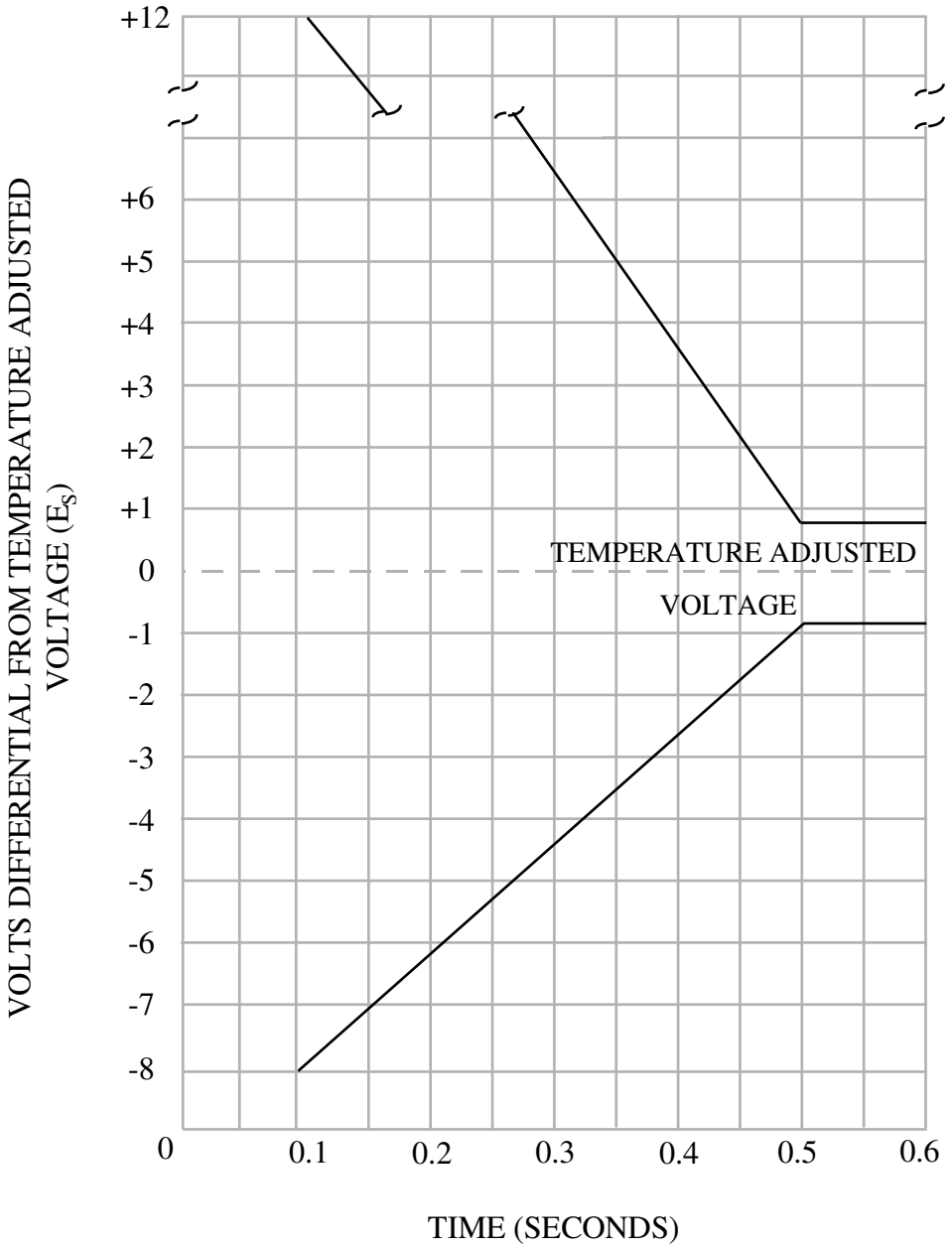


FIGURE 1. Voltage regulation.

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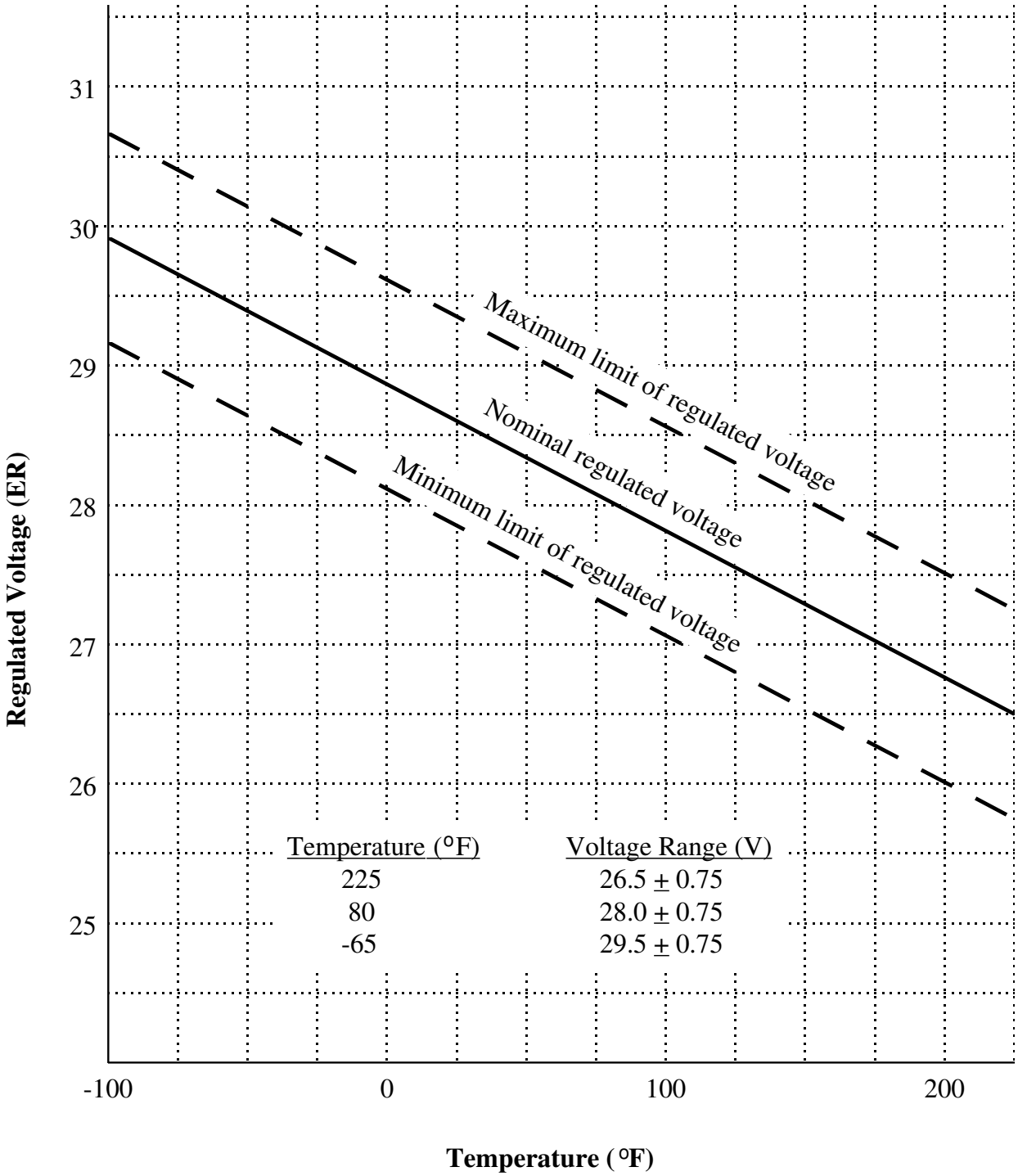


FIGURE 2. Voltage regulator temperature compensation.

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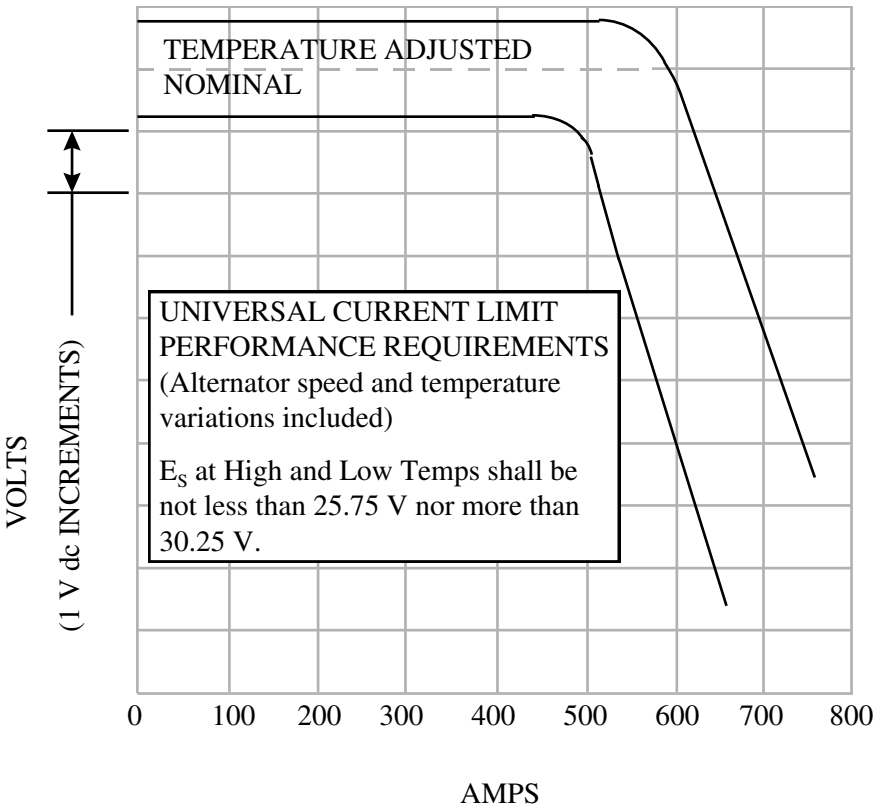


FIGURE 3. Universal current limit performance requirements.

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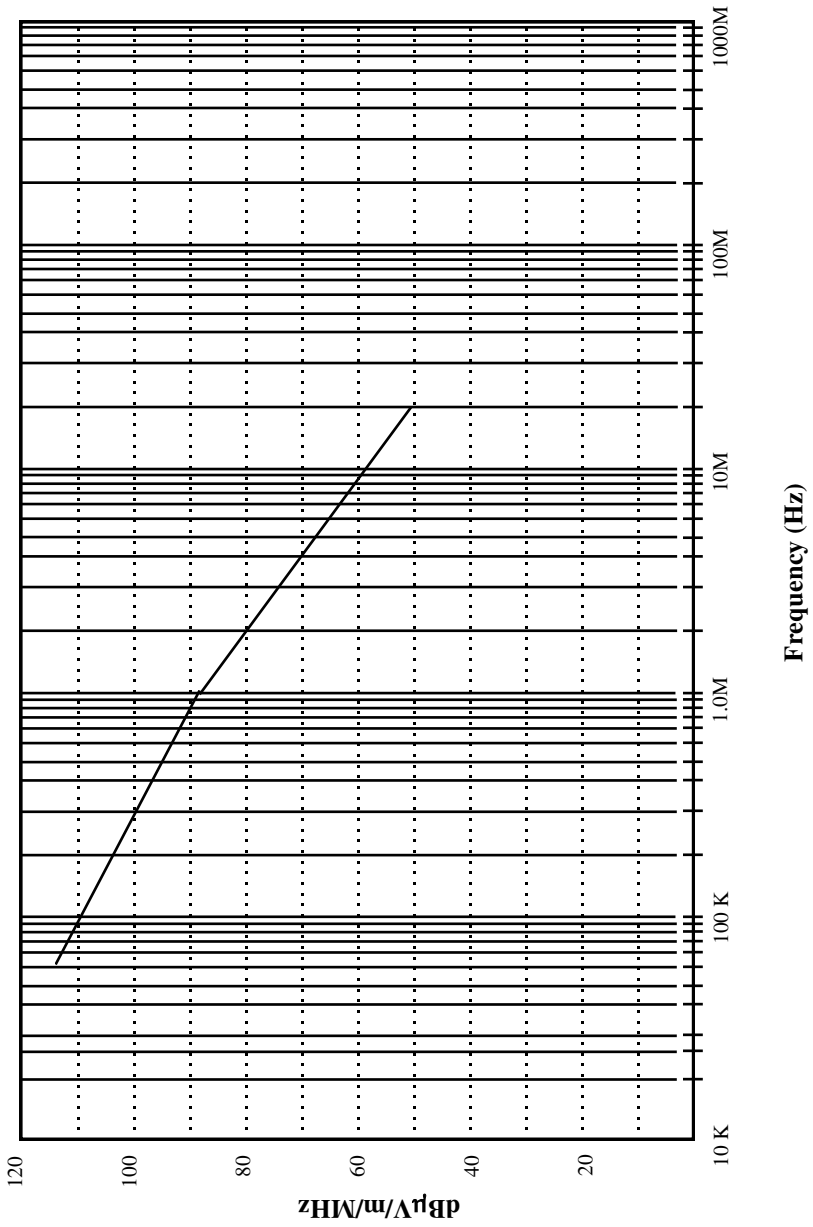


FIGURE 4. Broadband emission limits (conducted EMI).

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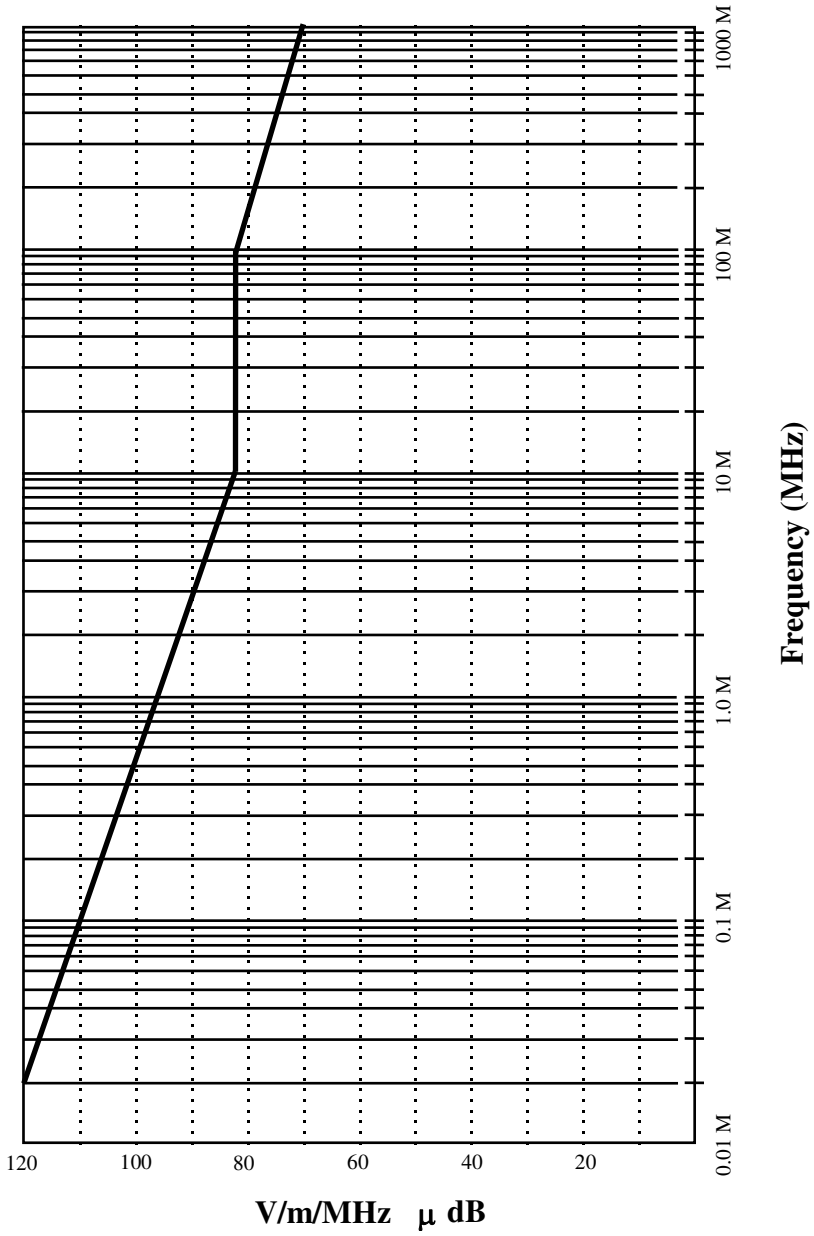
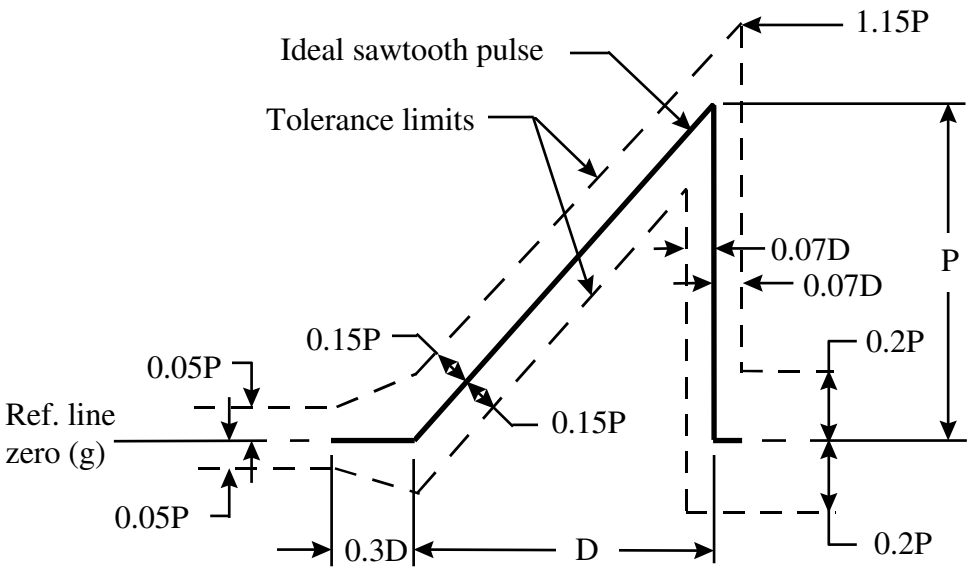


FIGURE 5. Broadband emission limits (radiated EMI).

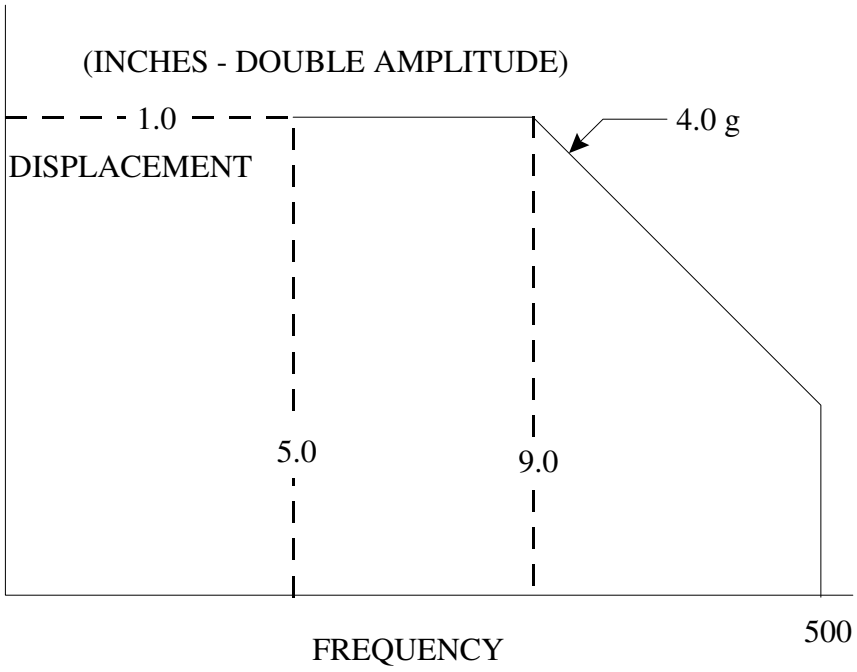
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NOTE: The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The peak acceleration magnitude of the sawtooth pulse is P and its duration is D. The measured acceleration pulse shock shall be contained between the broken line boundaries and the measured velocity change (which may be obtained by integration of the acceleration pulse) shall be within the limits $V_i \pm 0.1\ V_i$, where V_i is the velocity change associated with the ideal pulse which equals 0.5 DP. The integration to determine velocity change shall extend from 0.4D before the pulse to 0.1D after the pulse.

FIGURE 6. Sawtooth shock pulse configuration and tolerance limits.

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NOTE: ALL CURVES SHALL BE EXTENDED TO 2 CPS WHEN TEST ITEM RESONANCES BELOW 5 CPS ARE EXPECTED.

FIGURE 7. Vibration test curve for equipment installed in ground vehicles.

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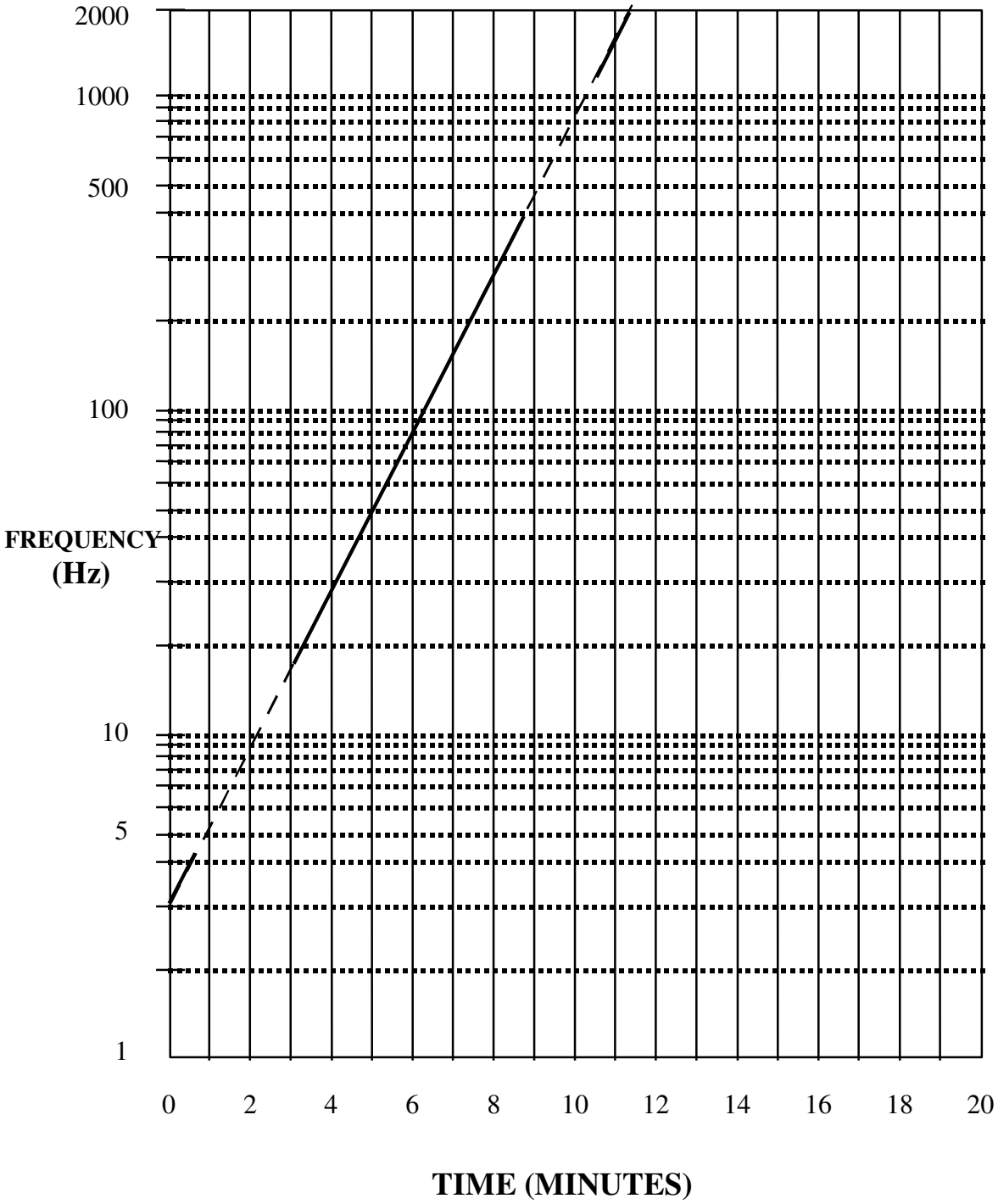
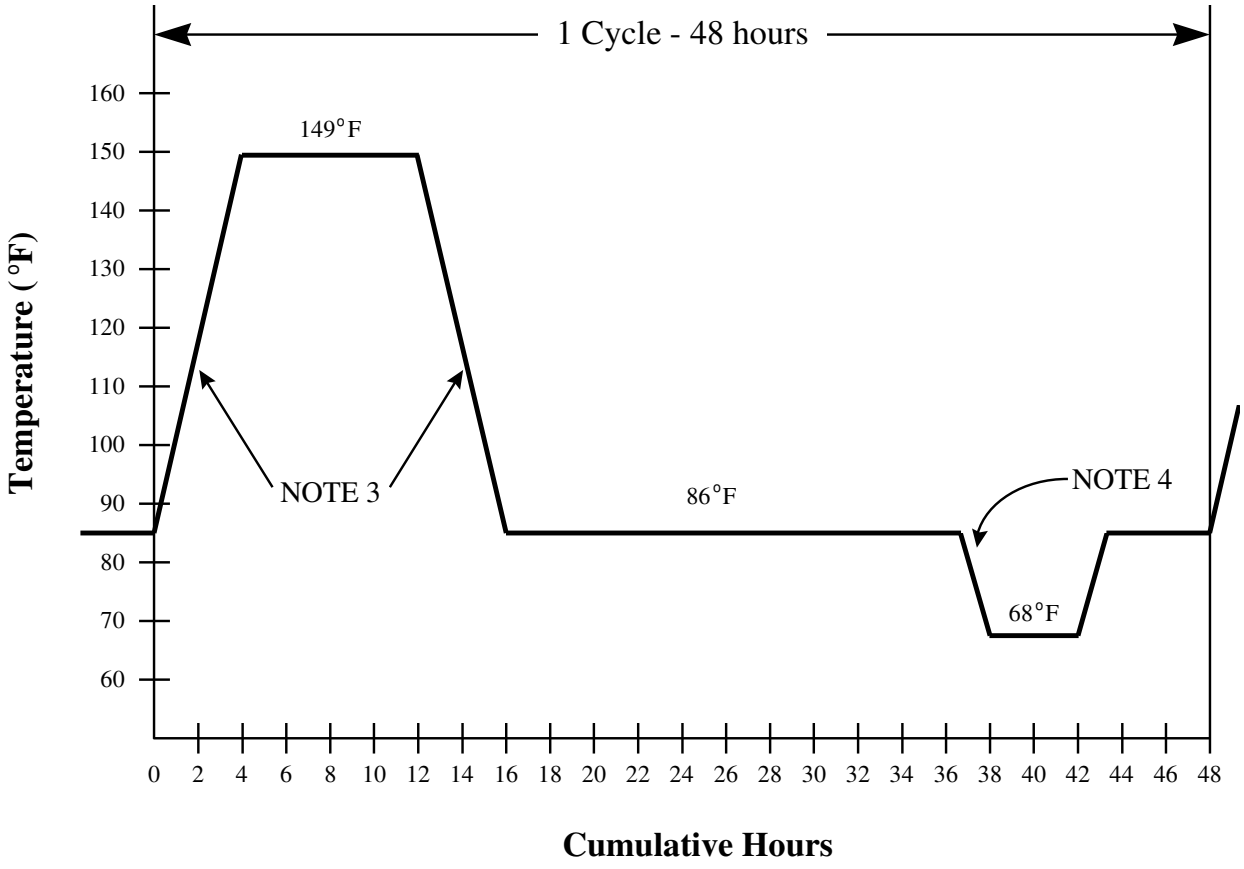


FIGURE 8. Logarithmic sweep.

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NOTES:

1. Tolerance during temperature change shall be not greater than 5°F.
2. Relative humidity shall be maintained at 94 $\pm$ 4% at all times, except that during the descending temperature period, the relative humidity may be permitted to drop as low as 85%.
3. Rate of temperature change between 86° and 149°F shall be not less than 14.4°F per hour.
4. The temperature change in this portion of the curve shall be not less than 18°F.

FIGURE 9. Humidity cycle.

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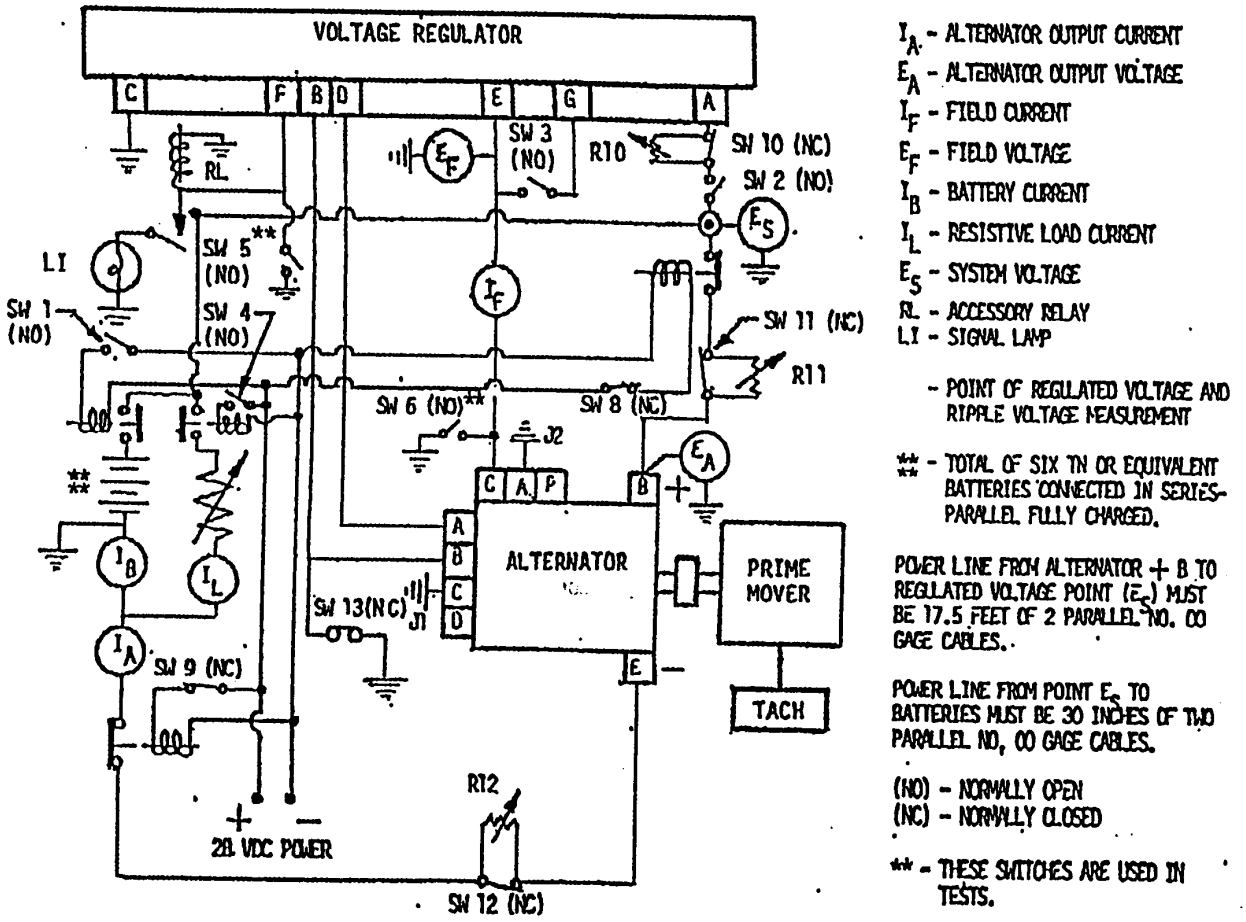


FIGURE 10. Voltage regulation test circuit.

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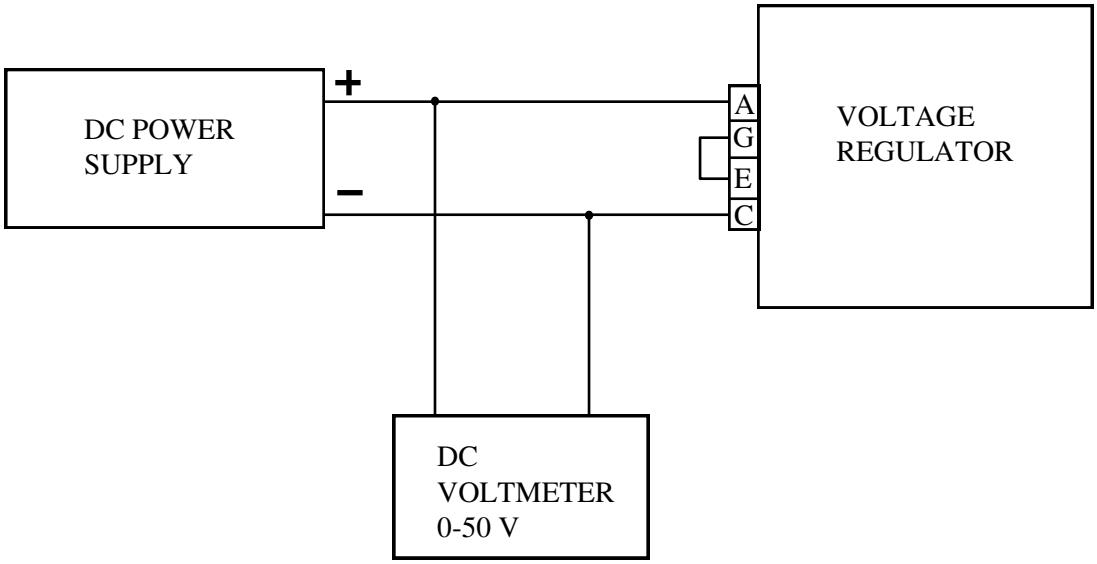
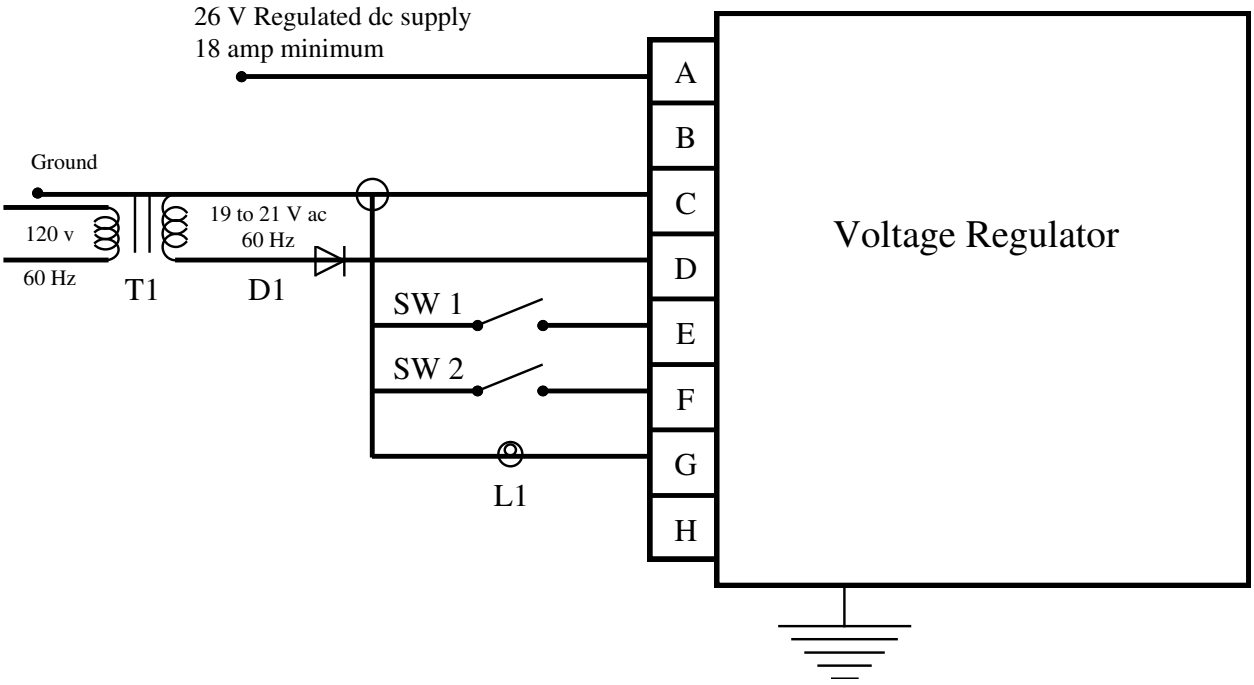


FIGURE 11. Overvoltage circuit test.

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NOTES:

SW1 - Normally open single pole-single throw (S.P.S.T), momentary, switch 18 amp minimum rating.
SW2 - Normally open S.P.S.T., momentary, switch, 1 amp minimum rating.
L1 - 28 Vdc lamp.
T1 - Transformer 60 Hz 19 to 21 V AC, 1 amp rating.
D1 - 3 amp, 50 volt, diode.

FIGURE 12. Short circuit protection.

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